

Advanced Tableau

Tableau Public overview

As you have been learning, Tableau is a powerful tool used by data professionals around the world to create, present, and share visualizations created from tabular data. If you have taken the Google Data Analytics Professional Certificate, then you should already be familiar with Tableau. If you haven't completed the Data Analytics Certificate, you can review resource materials below and linked in other videos. The Tableau software is available for free through its browser version, which allows learners like you to test out the capabilities of the software in a limited capacity. In this reading, you will be given an overview of Tableau Public, the free use, basic version of this visualization software.

Reviewing the fundamentals of Tableau Public

In this reading, you will learn about the basic structure of the **data source** and **design** screens featured in Tableau Public. The data source page is used for inputting or connecting to the data, and the design page is used for plotting and creating data visualizations. Both are needed to successfully design impactful and compelling data visualizations.

Note: To review the Tableau Public setup process, refer to the reading about [how to sign on to Tableau Public](#).

Data source page

Before you can start designing visualizations, you'll first need to upload your data. Since you've already set up your Tableau Public profile, all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar.

Note: Everything required for Tableau in this course can be completed with Web Authoring; you are *not* required to download the Tableau software.

Tableau Public Web Authoring

Web authoring allows you to create visualizations directly from a web browser. Can you create a viz without downloading any software? Yes! Since you've already [set up your Tableau Public profile](#), all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar. For the purposes of this

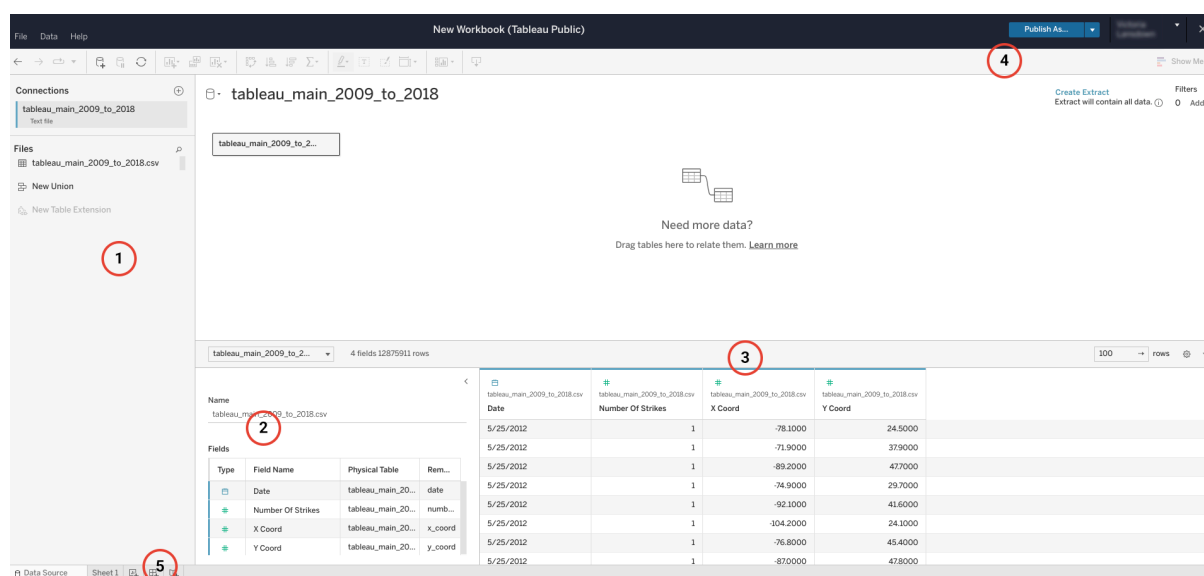
certificate program, you can perform everything you need in Tableau Public. The instructions in the following resources refer to Tableau Public.

Tableau Desktop Public Edition

You can also [download](#) the software directly to your Mac or PC. Select **Tableau Desktop Public Edition** under **Create** in the navigation bar on Public's website.

Reminder: Tableau Public should only be used for analyzing and sharing public data. All workbooks and datasets published will be freely accessible to anyone.

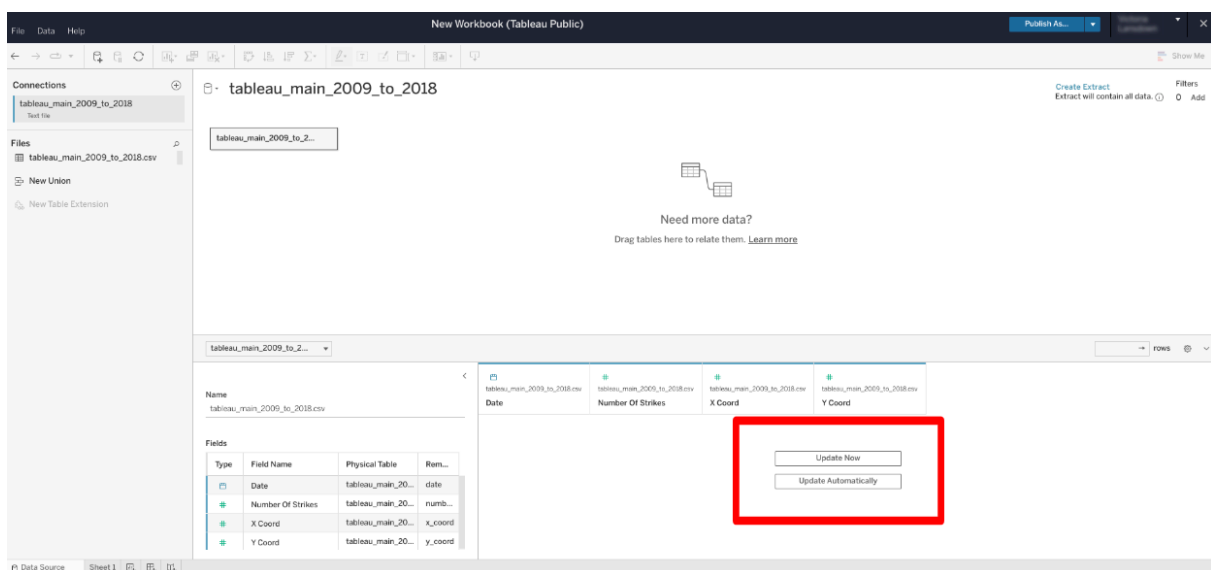
After you upload your dataset, you can perform the following steps outlined to match the circled numbers in the following image:



The following descriptions correspond to the image above.

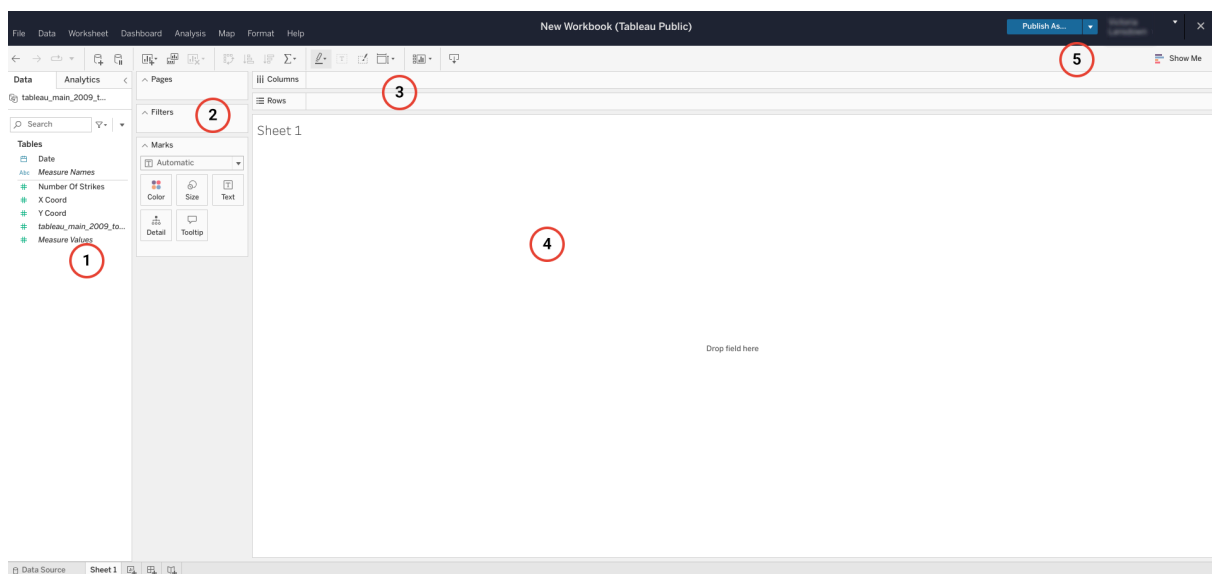
1. This left-hand pane includes your data connections and files. Here you will find all the files you upload in a list so that you can keep track of multiple files and/or multiple connections to different databases.
2. Just to the right of the data connections window is a list of all of the fields that Tableau Public has detected in a particular file. If you have multiple files uploaded, you can select the file from a dropdown to access each file's fields. As you'll learn in an upcoming video, Tableau's fields are acquired from the data columns in your file. Tableau automatically sorts these fields into dimensions or measures and discrete or continuous variables.

3. The biggest pane on the page, on the middle right, allows you to access all of the columns of your file as Tableau fields, including several rows of data. Unlike the pane to the left, this pane allows you to create new fields based on those already present, such as new calculation fields, groups, sets, or parameters (you will learn more about these features in upcoming videos). You may be prompted to select "update now" or "update automatically" in order to populate this pane with your data. If that's the case, it is good practice to update automatically to ensure you're consistently working with recent data. (For reference -- review the following image, after #5.)
4. The blue button "Publish" in the upper right of the screen acts as your "save" button. Because Tableau Public is a browser-based platform, anything you create and want to save will be published to your public account. There are ways to password lock or hide data sources and data visualizations if desired, but Tableau Public only offers the Publish field for saving your work. When you click on the 'Publish' button, you may automatically be navigated to your data design page, which may be blank depending on your design progress. Do not be alarmed. Your latest dataset uploads or data designs were still saved; just navigate back to where you were last and continue editing your visualization.
5. Lastly, you will use the collection of buttons at the bottom left of the page to navigate to your data design page. You'll find button options for creating a new worksheet, a new dashboard, and a new story. These elements will be introduced in the next section.



Data design page

The data design page is where your data visualizations will be built. To navigate to the data design page, click on 'Sheet 1' or create a new sheet as instructed in #5 on the previous corresponding image. When you first click to open a data design page, you may be prompted that Tableau is 'Creating Extract'. That means that Tableau is extracting the providing data to be used in visualizations. This process may take several minutes. Here you will move your data source fields to appropriate shelves to build the type of visualization you want. You can build data visualizations or entire interactive dashboards from this page.



The following numbered items correspond to the numbers displayed in the Tableau workbook image above.

1. In this pane on the far left, you will find your list of discrete and continuous dimensions and measures. You will move these variables to different panes on this page to build visualizations. You will learn more about these variables later.
2. In the next pane just to the right, you'll find "Pages," "Filters," and "Marks." You can move any dimension or measure to these different fields to manipulate the data visualization. You will learn how to use these features in upcoming videos.
3. At the top of the page, just under the menu bar, there are two empty rows that act as your main two shelves for moving your variable fields. The "Columns" and "Rows" shelves help you position your data visualization as

desired. You'll also notice above these rows a toolbar and menu full of other options for manipulating your data visualization.

4. In the middle of the screen is the main viewing panel for your visualization. As you add elements and drag your dimensions and measures to different fields, you will notice the impact they have on your data visualization in this panel. In the upper right corner you will find your "Publish" button, which acts as the save button, and the "Show Me" dropdown. Under the "Show Me" dropdown, you will find a selection of data visualization types and guides for building each of them.
5. When you're ready to save and share your work, publish it to your Tableau Public profile. View the options for publishing your work by clicking on the down arrow next to the "Publish" button on the top navigation bar.

Other ways to review Tableau tools

Google Data Analytics Professional Certificate

As mentioned earlier, if you have taken the Google Data Analytics Professional Certificate program, you are already familiar with Tableau and Tableau Public. Go to the Get started with Tableau lesson in that program to review what you have learned.

Tableau.com

By visiting Tableau.com, you'll notice multiple product offerings, everything from Tableau Public (which is free) to Tableau Desktop, Tableau Mobile, and Tableau Server. Each product has its own use and specialization, but the main elements for data visualization are the same. You can search through the Tableau Help page to find specific articles on just about any topic regarding data visualizations. There are a variety of training resources available from Tableau to help users learn the different features of their products. Tableau offers a variety of training resources for helping to learn their different products.

Key takeaways

Tableau is a powerful data visualization tool, but that means it takes a lot of practice and experience to use it proficiently. The two main pages you'll use are the data source and data design pages. There are also a large number of

resources available to help you in each step of the process, including Tableau Help and Grow With Google Data Analytics Certificate Program.

Resources for more information

To help you troubleshoot or to learn more, you can use the following links:

- Use Tableau resource page to set up your data for success: [Set up data sources](#)
- Tableau Tools and Web Authoring Help: [Design charts and analyze data](#)
- The Tableau Public “Discover” page, which includes “Viz of the Day” and other beautiful vizzes designed on the platform: [Welcome to Tableau Public](#)
- Beginner's guide to using Tableau Public: [A step-by-step guide to get you started on your own data viz journey](#)
- Getting Ready to Publish Your First Data Visualization: [A step-by-step guide for analyzing data and publishing a viz to your Tableau Public profile](#)

How to sign on to Tableau Public

As you have been learning, Tableau is a powerful data visualization tool used by data professionals around the world. Tableau offers a free browser version of its software, Tableau Public, which allows learners like yourself to test out its capability in a limited capacity. Tableau Public is the software you will use throughout this course on data visualization. In this reading, you will be guided on how to sign up for Tableau Public, and you will be given troubleshooting resources for potential obstacles you may experience.

How to sign in to Tableau Public

Signing in with an existing Tableau.com account

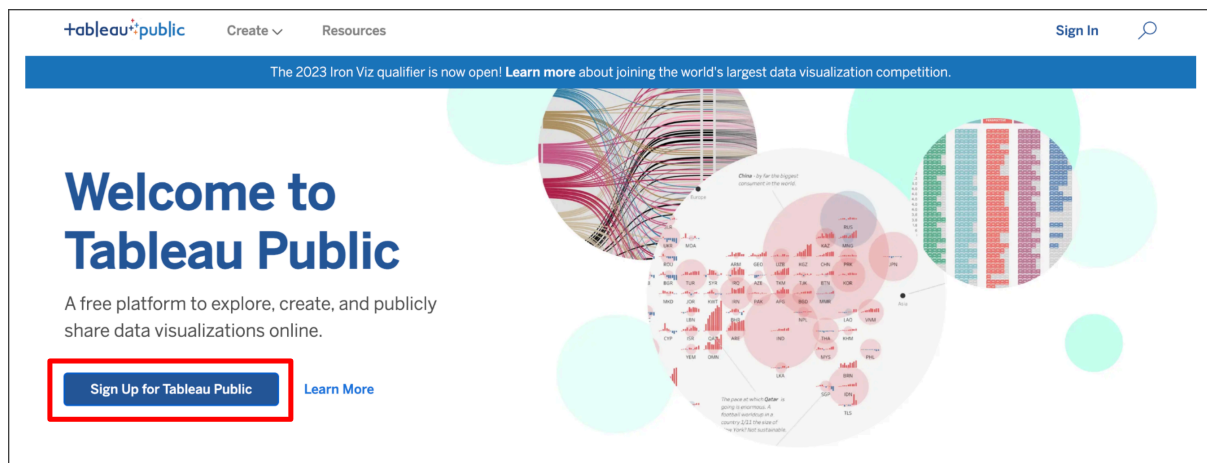
If you already have a Tableau.com account, you can use your existing [login credentials](#) to sign into Tableau Public. Click [here](#) to learn more. If this is your first time signing in to Tableau Public with your Tableau.com account, you can set your account password by resetting your password. To do so, click “[reset password](#).”

Signing in for the first time

The following instructions show you how to sign up for Tableau Public for the first time, without a pre-existing Tableau.com or Tableau Public account. First, go to the [Tableau Public home page](#). If [this link](#) does not open to Tableau's homepage, type public.tableau.com in your browser's address bar.

Note: *Tableau Public works best on Chrome (Windows, Mac, Android), Edge (Windows), Firefox (Windows and Mac), Safari (Mac and IOS).*

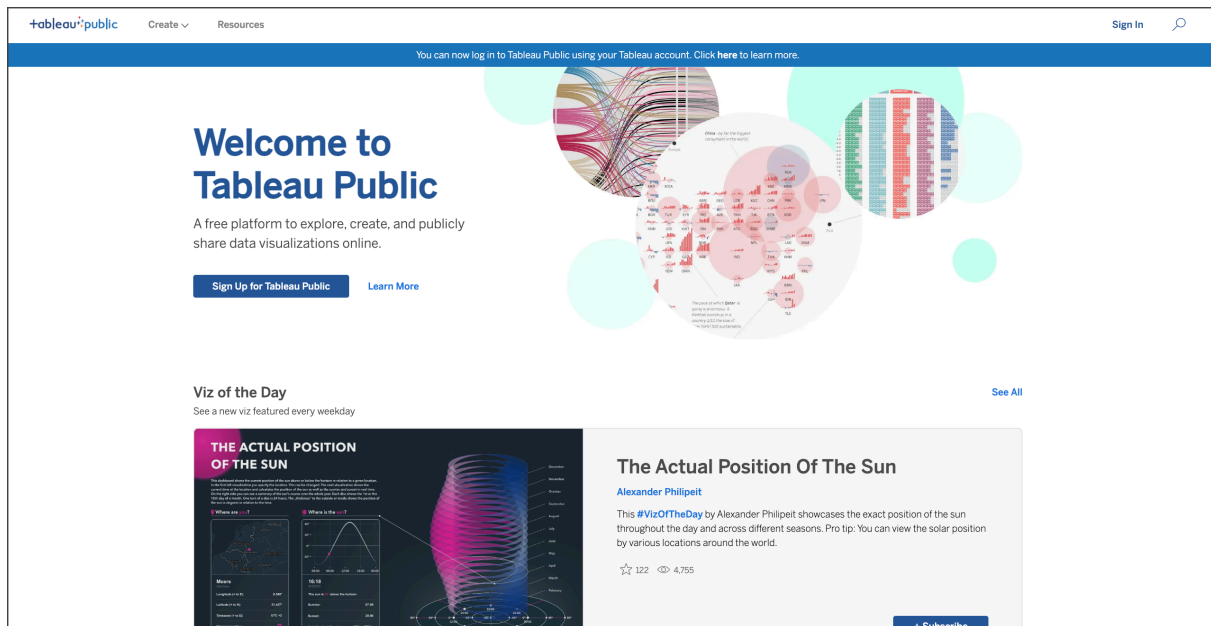
Next, at the top right of the page, click the "Sign Up" button.



Fill in all the required fields and click "Create My Account."

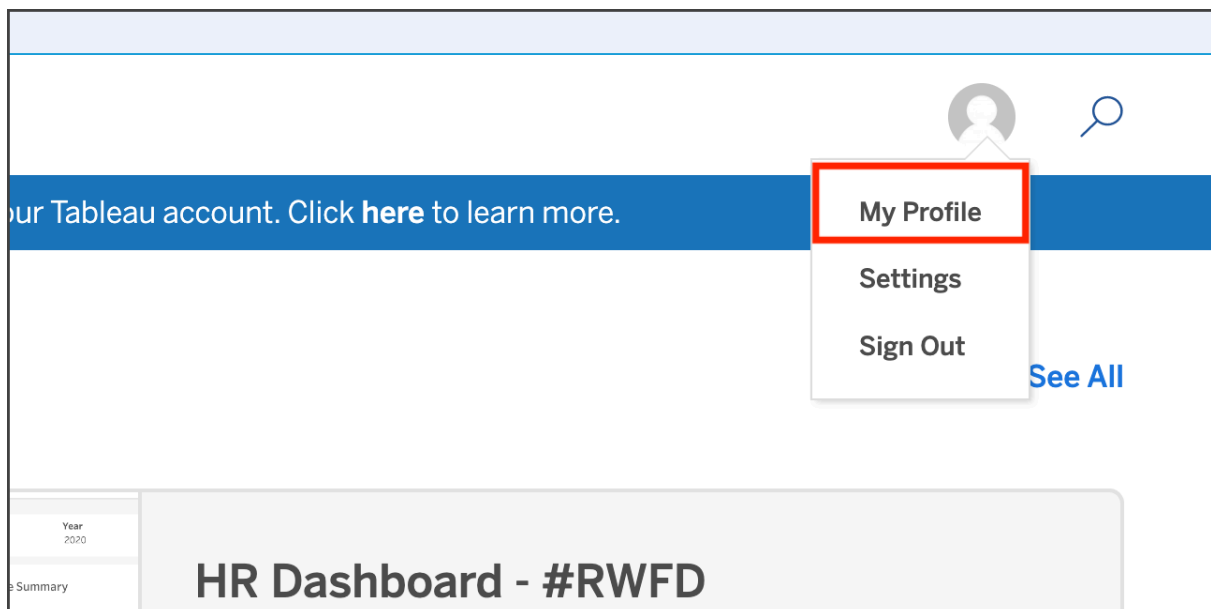
Navigating Tableau Public

The home page for Tableau Public contains resources and guides for helping data professionals learn more about and get inspired by data visualizations. You can explore any and all of the links on this landing page to help enhance your knowledge of Tableau. The most helpful place to start if you are new to Tableau is the [Resources page](#), which has how-to videos, community resources and user forums.



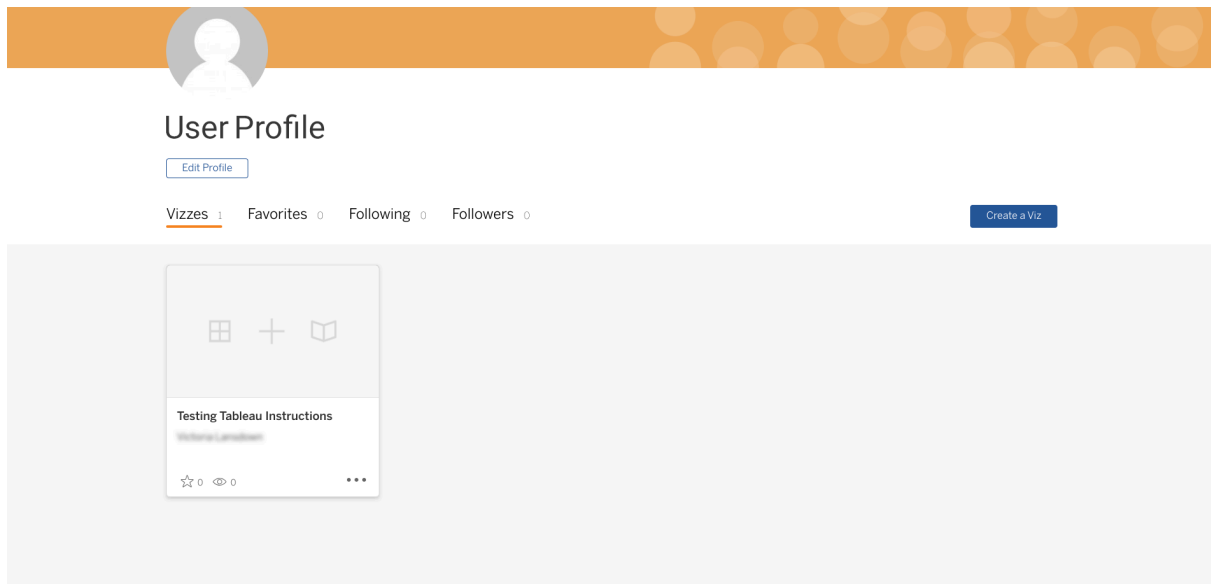
How to start creating your own visualizations

During the instruction videos, the instructor will ask you to log in to Public Tableau and follow along in the creation of a data visualization. Once you've set up your Tableau Public profile, all you need to do is log in and select Web Authoring under Create in the navigation bar.



Get Started

Within your Tableau Public profile, you'll find tabs for *Vizzes*, *Favorites*, *Following*, and *Followers*. At the instructor's prompt, click *Create a Viz*.



Key takeaways

Accessing Tableau Public and creating a profile will be your first step to learning how to design data visualizations like a data professional.

Resources for more information

To help you troubleshoot or to learn more, explore take a look at the following links:

- Tableau Public not working? Check out these [Technical speculations and storage requirements](#)
- [The Tableau Public Discover page](#) includes 'Viz of the Day' and other beautiful vizzes designed on the platform
- [The Tableau Public Blog](#) offers regular dispatches from the Tableau Public Team

Download your datasets and begin presenting with Tableau

Accessing and utilizing resources in this section

To help your learning in this part of the course, follow along with the instructor as they go through Tableau in each video.

Steps to complete:

1. Open [Tableau Public](#).
2. Since you've already set up your Tableau Public profile, just log in and select Web Authoring under Create in the navigation bar.
3. Download the dataset files attached below and then upload them to Tableau. ***See additional instructions below.**
4. Open the Tableau follow-along guides linked below. These guides are labeled to correspond to video titles and concepts.
 - [Tableau follow-along_guide: Work with Tableau, Part 1](#)
 - [Tableau follow-along_guide: Work with Tableau, Part 2](#)
 - [Tableau follow-along_guide: Craft compelling stories with Tableau](#)
 - [Tableau follow-along_guide: Present like a pro with Tableau](#)
4. Follow along with the instructor in each video as they show the creation of the data visualizations in Tableau.

In this lesson's videos, you will need access to these datasets:

- **Instructions for downloading the dataset files and uploading them to Tableau:**

You will need to download the provided .csv files and upload them to Tableau Public to follow along with the instructor in subsequent videos and interact with Tableau. To do so, follow these steps:

1. Download the attached .csv zip file to your device.
2. Extract the contents of the zip file once the download is finished.
 - a. You'll need to upload the specific dataset files to Tableau. Do not upload the entire .zip folder.
 - b. When you download the zip folder from this page, your computer will automatically download a .zip file folder. The .zip folder is automatically named with a series of letters and numbers.
 - c. Open up that .zip folder, then save the individual dataset files. The two files are: tableau_main_2009_to_2018.csv and tableau_dataset.csv.
 - d. Once you can see the individual dataset files, proceed to upload those to Tableau Public.
3. Upload the extracted .csv files to Tableau.

- a. Review the provided [Tableau follow-along guide: Work with Tableau, Part 1](#), and/or follow along with the instructor in the video [Work with Tableau, Part 1](#) for further instructions about uploading files to Tableau.
- b. There are two Tableau datasets: tableau_main_2009_to_2018.csv and tableau_dataset.csv. Be sure to check the dataset name shown on screen in the instructional video.
- c. When following along, make sure you've uploaded the same file so you can create the same visualizations.

Data dictionary

The datasets above represent lightning strike counts in the United States. They include four columns of data: latitude, longitude, date, and lightning strike counts. Each row represents a total lightning strike count on the specified date for a particular location.

Column name	Type	Description
number of strikes	int64	The total count of lightning strikes on a given day
X Coor	obj	Longitude
Y Coor	obj	Latitude
date	str	The recorded date (format: DD/MM/YYYY)

Follow-along guide: Work with Tableau, Part 1

This document includes detailed instructions for how to perform the data visualizations described in the video "Work with Tableau, Part 1."

The following guide points out areas of the video that may require adjustment. These reference guides can also serve as a set of usability reminders for you to recall when using Tableau in your future career.

Instructions

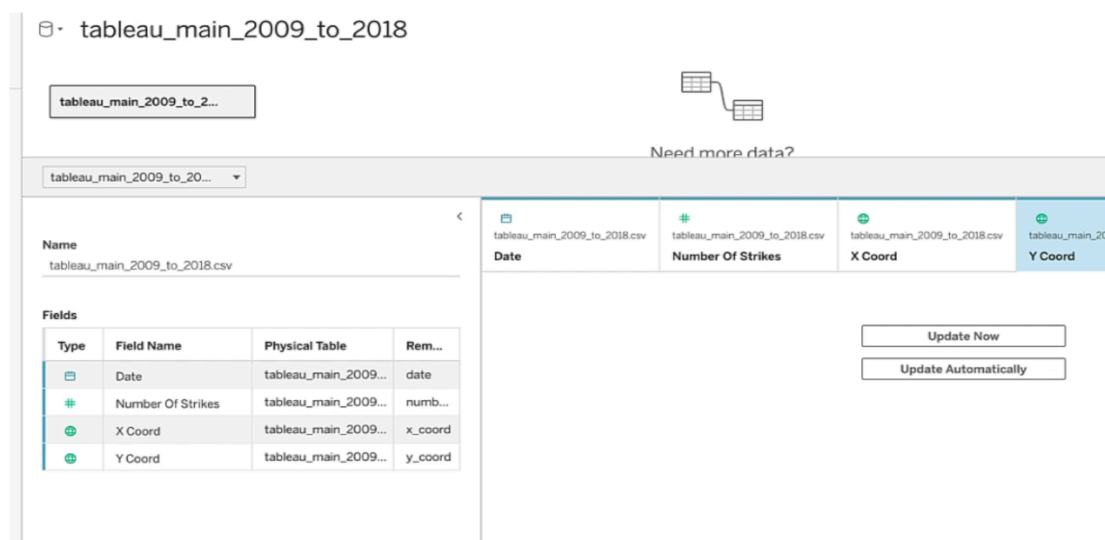
- Since you've already [set up your Tableau Public profile](#), all you need to do is log in and select **Web Authoring** under **Create** in the navigation bar.
- **Upload your dataset from your computer.** Select the appropriate CSV file provided in the [instructions](#). The dataset you'll use with this instructional

video is: tableau_main_2009_to_2018.csv.

(Note: Please allow several minutes for data upload into Tableau Public.)

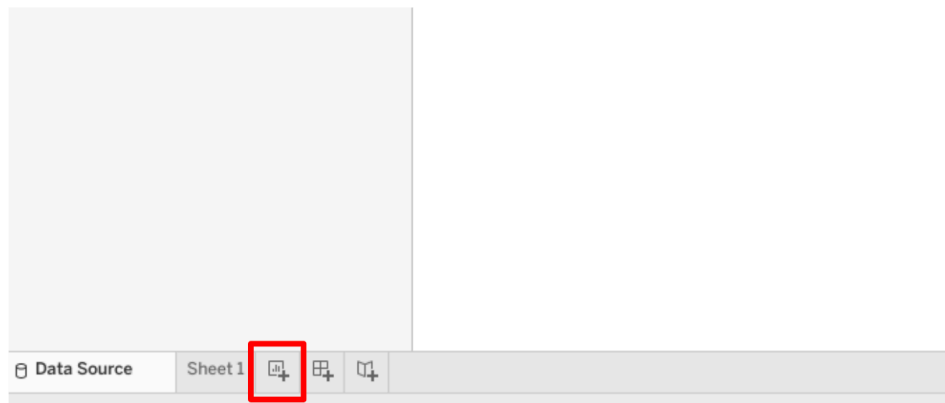
Before you can start designing visualizations, you'll first need to upload your data. You'll need to upload the specific dataset files to Tableau. Do not upload the entire .zip folder. When you download the zip folder from this page, your computer will automatically download a .zip file folder. The .zip folder is automatically named with a series of letters and numbers. Open up that .zip folder, then save the individual dataset files. The two files are: tableau_main_2009_to_2018.csv and tableau_dataset.csv. Once you can see the individual dataset files, proceed to upload your dataset for this video to Tableau Public.

Notice on the data source tab that you can see all of your column headers and Tableau icons that help you determine data types. In this case, you'll see a calendar icon and pound signs indicating numbers or integers.



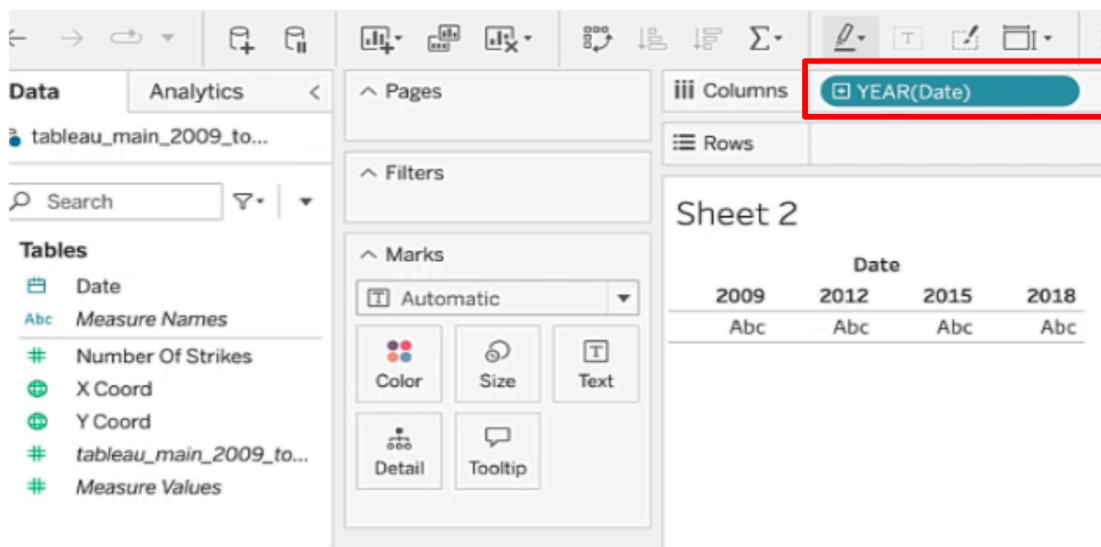
- Click on **New Worksheet**.

(Note: Please allow several minutes for data to import into a new worksheet.)

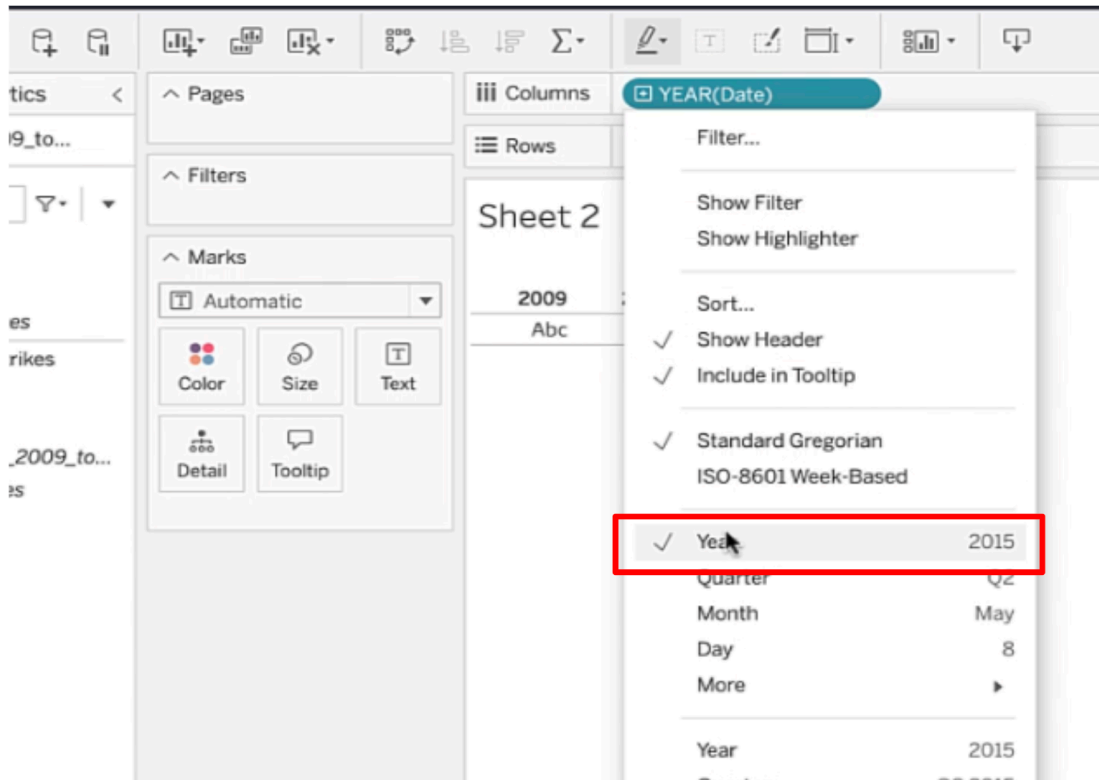


Notice the green and blue fields and the gray line dividing them. As you'll remember from the video, blue indicates a discrete field and green indicates a continuous field. The gray line divides the dimensions from the measures, with dimensions above and measures below the line.

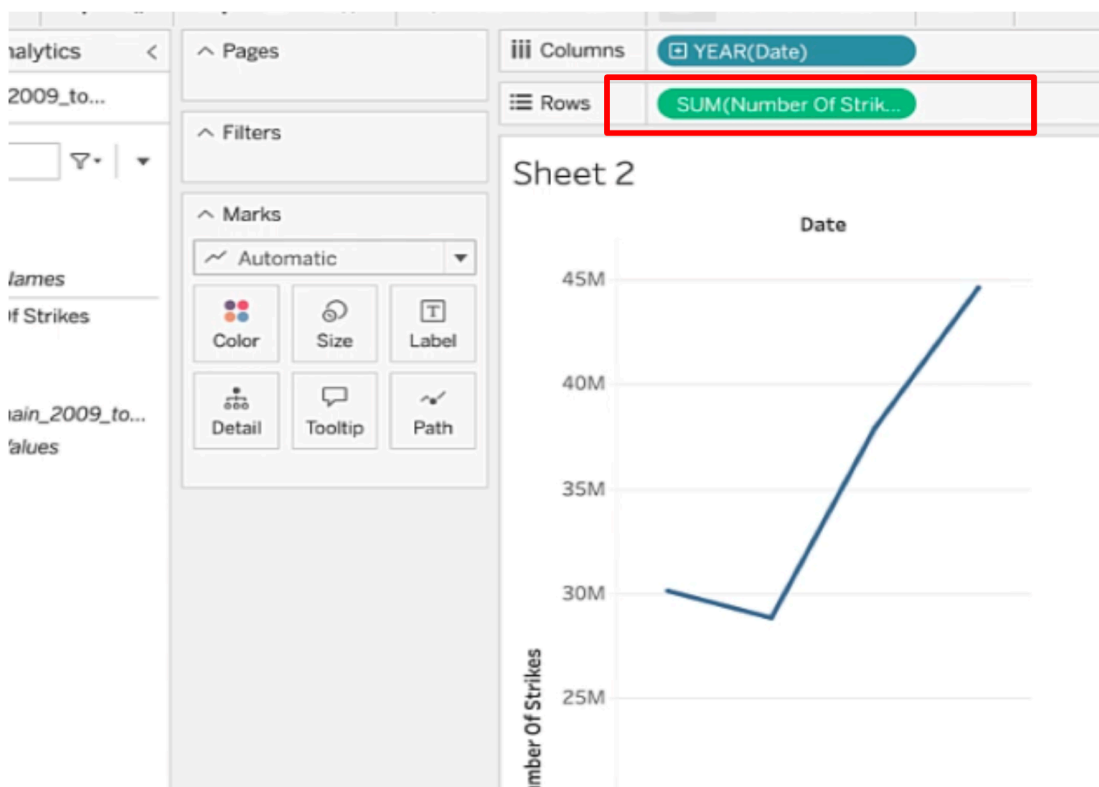
- Drag **Date** into Columns shelf.



- Click on the drop down menu of the date field. Ensure there is a checkmark next to **Year**.

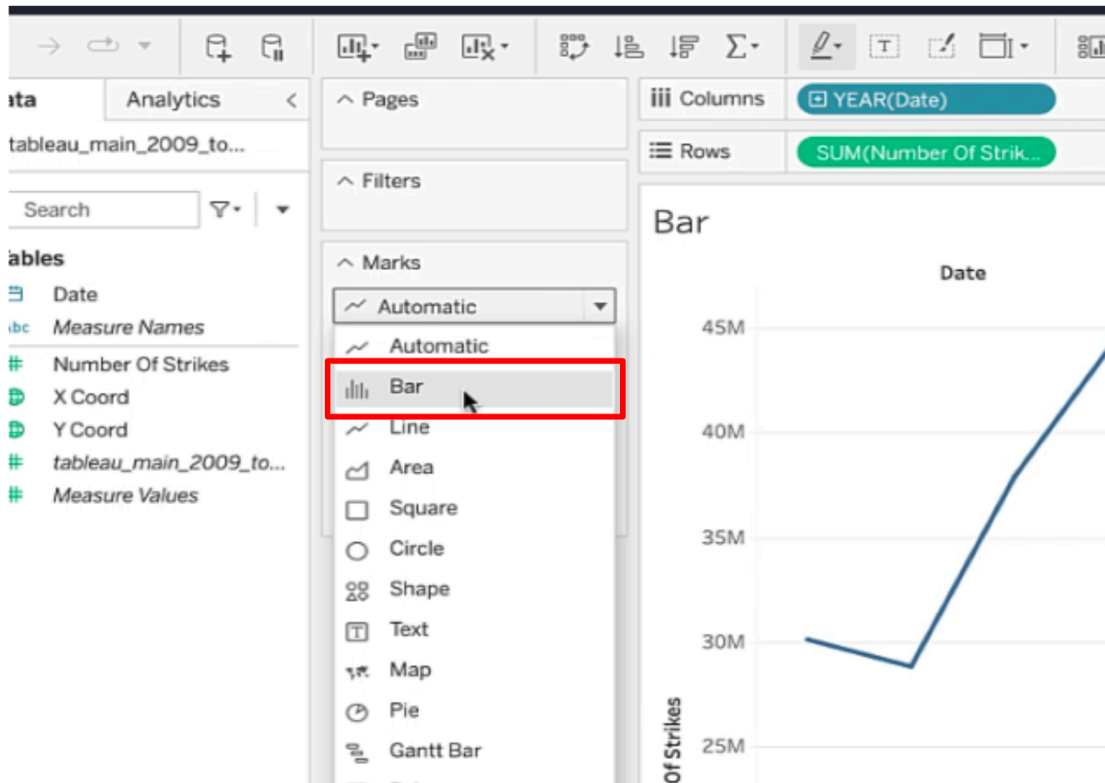


- Drag **Number Of Strikes** to the Row shelf. You've created a line graph.

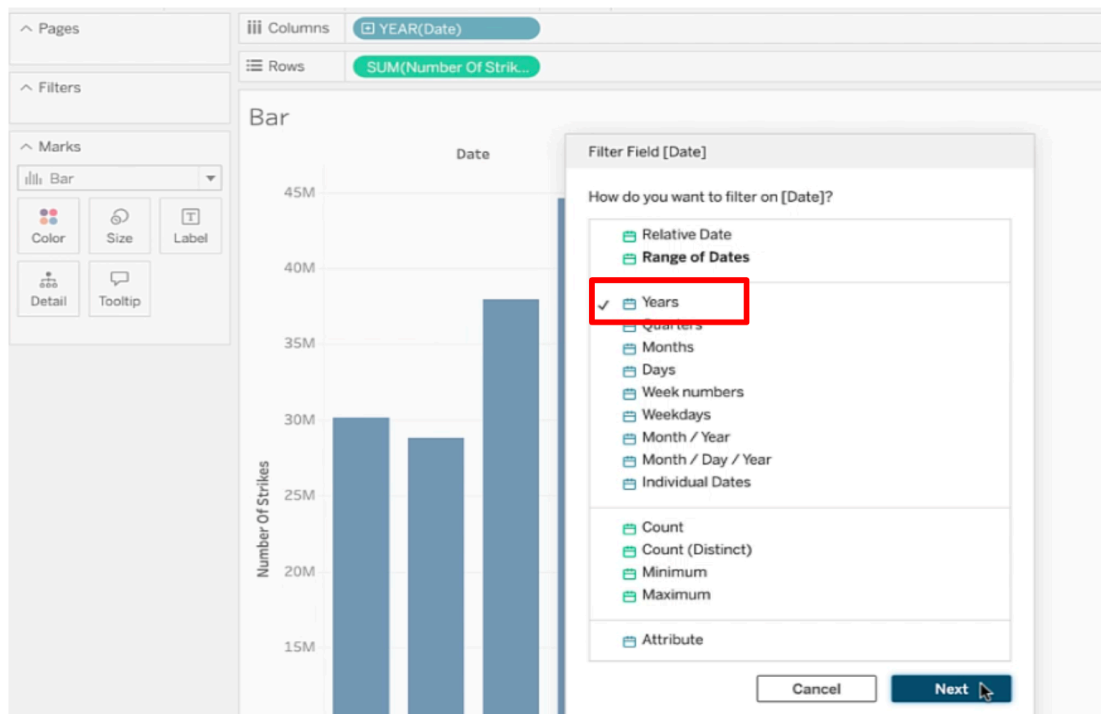


- Click on **Duplicate** worksheet in the toolbar menu.

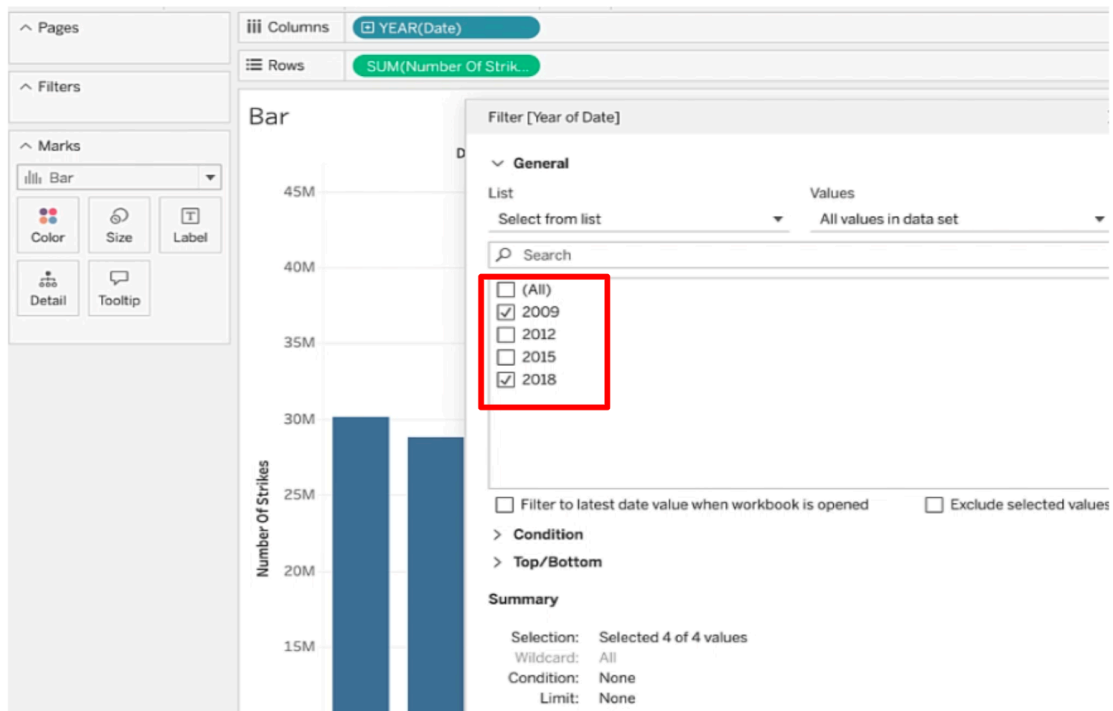
- Click on the dropdown menu in the **Marks** field. Select **Bar**.



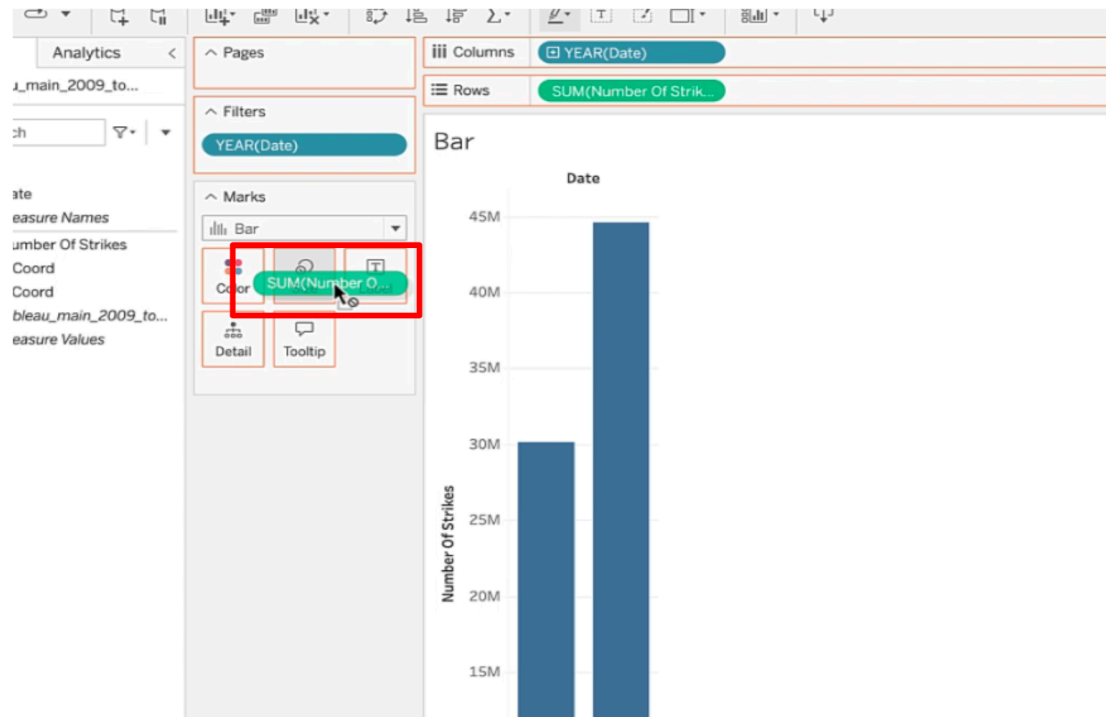
- Click **Filters**.
- Select **Years** when the pop-up window appears.
- Click **Next**.



- Select only **2009** and **2018** in the next pop-up window.
- Click **OK**.

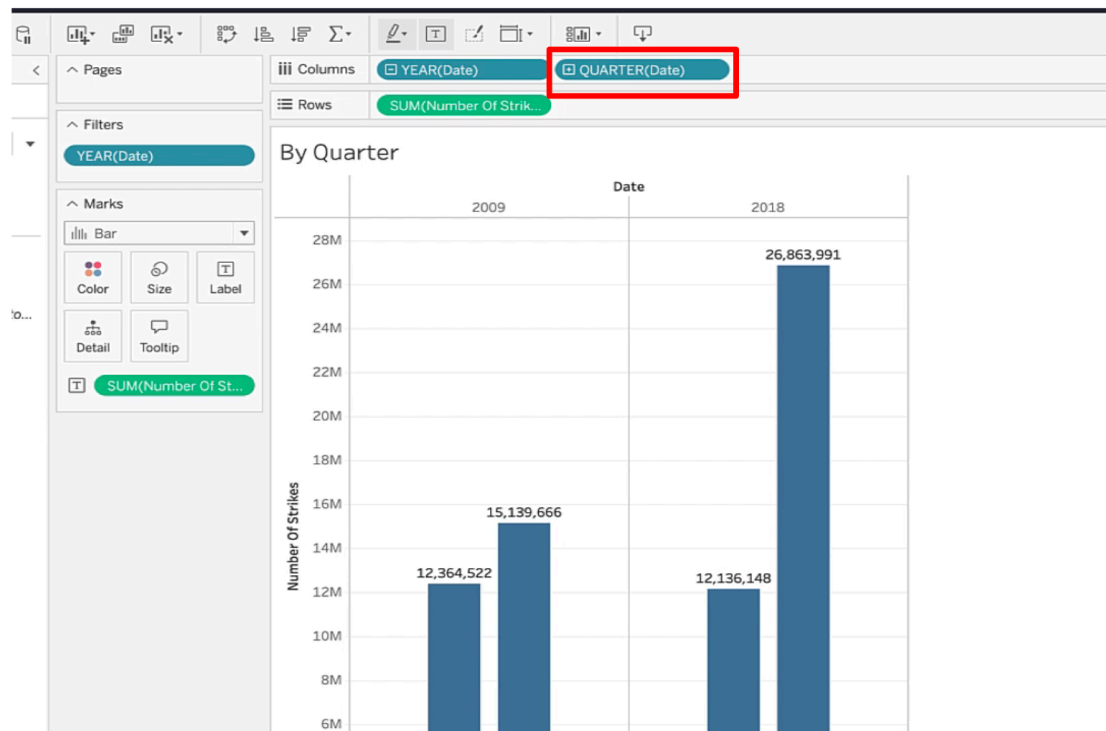


- Add labels to the bars by dragging **Number Of Strikes** to the **Label** field.

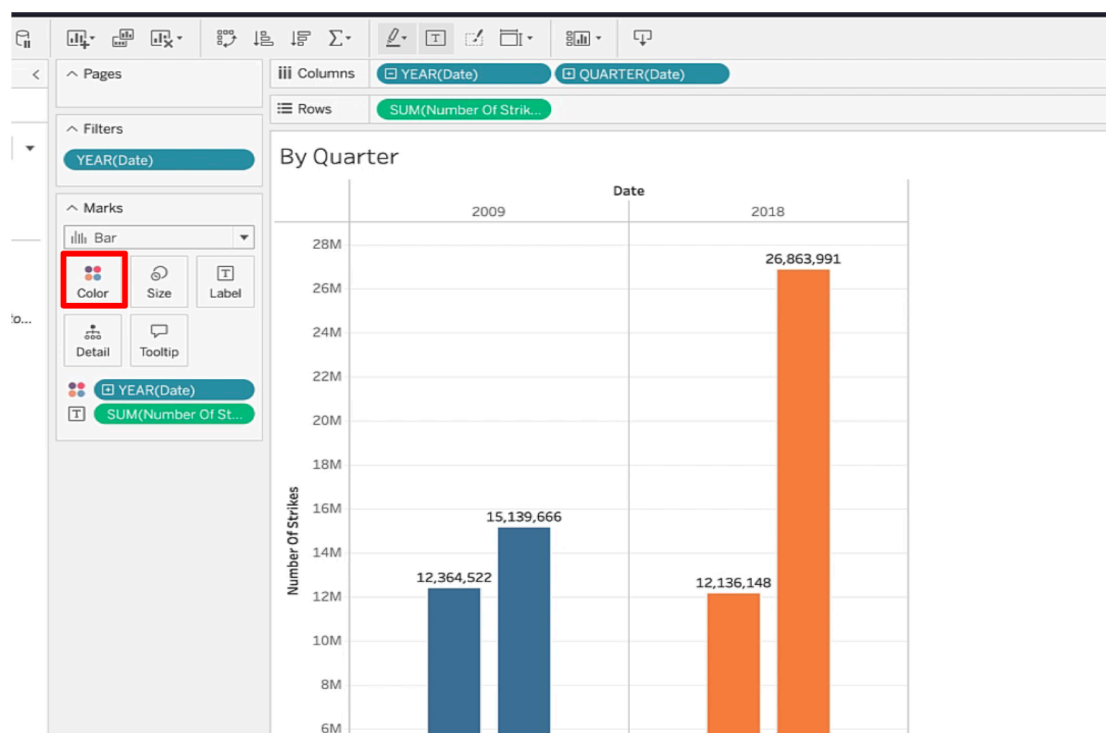


Notice the bars are labeled at the top with total number of strikes for 2009 and 2018.

- Click **Duplicate** worksheet in the toolbar menu.
- Drag **Date** to column shelf. This will automatically divide the number of strikes into quarters.



- Drag **Date** to **Color** square.



- Click on the blue drop down arrow of **YEAR(Date)** in the **Color** field and select **Quarter**.

You're done!

(**Note:** Be sure to save your work to your Tableau Public profile by clicking on "Publish.")

